

GEORGE JOBI PERANGATTU

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PROFESSIONAL SUMMARY

Robotics engineer pursuing an M.S. in Robotics and Autonomous Systems with a B.S. in Computer Science and hands-on experience across robot navigation, ROS/ROS 2/ROS2, C++, Python, computer vision, sensor fusion, controls, controllers, and real-hardware deployment. Experienced building and evaluating robotics algorithms on physical hardware and in CARLA/Gazebo simulation, with research in camera-LiDAR fusion, robot learning pipelines, model deployment constraints, edge compute considerations, and GPU-accelerated PyTorch training. Strong fit for physical AI field deployment, robot integration, sensor suites, model evaluation, experiment design, data collection, and translating field learnings into scalable robotics capabilities.

EDUCATION

M.S. Robotics and Autonomous Systems, AI Concentration, Arizona State University, Tempe, AZ - Graduating Jul 2026

B.S. Computer Science, Arizona State University, Tempe, AZ - Graduated May 2025

TECHNICAL SKILLS

Robotics: ROS/ROS 2/ROS2, robot navigation, sensor fusion, camera-LiDAR fusion, integrated sensor suites, localization, motion planning, RRT, trajectory optimization, collision avoidance, kinematics, PID control, closed-loop control, controllers, robot integration, field evaluation

Machine Learning and Vision: PyTorch, TensorFlow, transformers, CNNs, computer vision, object detection, segmentation, tracking, pose estimation, depth estimation, BEV perception, 3D point clouds, LiDAR, RGB, RGB-D cameras

Simulation and Deployment: CARLA, Gazebo, synthetic data generation, URDF, simulation-to-real validation, experimental design, model evaluation, data collection pipelines, GPU training, mixed precision, bandwidth/compute/latency tradeoffs

Languages and Systems: Python, C++, C, Java, JavaScript, TypeScript, SQL, Linux/Ubuntu, Docker, Git, CMake, CI/CD, GDB, testing, debugging, technical documentation

ROBOTICS AND ML PROJECTS

Efficient TransFuser: Multi-Task Sensor Fusion for Autonomous Navigation, ASU CSE 598 Research Project - Spring 2026

- Designed a camera-LiDAR sensor fusion architecture for autonomous navigation, replacing TransFuser's RegNetY backbone with EfficientNetV2-S to reduce parameters by 70% while improving held-out loss by 8.7%.
- Developed multi-scale transformer fusion for RGB and LiDAR point cloud processing, improving BEV segmentation by 25.4% across navigation, semantic segmentation, detection, depth estimation, and BEV mapping tasks.
- Built data loading and training pipelines for multi-task robotics perception in CARLA, using synthetic data generation, controlled experiments, and reusable evaluation code for model training and validation.
- Diagnosed AMP instability in CenterNet focal loss during FP16 GPU training, showing ability to reason about model architecture, latency, compute, and systems constraints when choosing deployment approaches.

Parrot MiniDrone: Autonomous Robot Navigation on Real Hardware, ASU Robotics Coursework - Spring 2026

- Implemented real-time autonomous navigation on a physical aerial robot using closed-loop PID control, inertial and visual sensor fusion, and model-inference-driven test sequences.
- Logged and analyzed sensor data to evaluate robot behavior, debug field issues, and improve reliability under real-world constraints.
- Built safety checks for autonomous operation and validated navigation algorithms across simulation and real hardware.

3D Motion Planning for Robot Navigation in Gazebo, ASU Robotics Coursework - Fall 2025

- Developed Python-based 3D RRT motion planning for collision-free robot trajectories and integrated experiments with ROS and Gazebo.
- Applied trajectory post-processing and ran repeatable evaluations across multiple environment configurations.

Spider CAD Robot: Multi-Legged Robot Design, ASU Robotics Coursework - Spring 2026

- Designed a multi-legged robot in CAD with constrained joints, validating kinematics and preparing a URDF-exportable model for ROS/Gazebo simulation and controls experimentation.

Factory Automation System for Semiconductor Process Simulation, ASU Applied Project - Starting Jun 2026

- Planned industrial robotics and automation system using LiDAR, cameras, actuators, control logic, and sensor-fusion-driven pipelines for manufacturing-process simulation.

WORK EXPERIENCE

Backend Developer Intern, DigiClips Media, Tempe, AZ - Aug 2024 to May 2025

- Developed ML models for image recognition using Python and PyTorch, optimizing data loading and GPU-aware training pipelines handling 10M+ daily queries and reducing latency by 45%.
- Improved computer vision detection accuracy by 30% through reusable code, evaluation loops, and cross-functional iteration.

Full Stack Developer Intern, Odoo, Dubai, UAE - May 2025 to Aug 2025

- Refactored Python backend systems for production applications and collaborated on deployment-oriented engineering work.

Full Stack Developer Intern, TicketDex, Tempe, AZ - Jan 2025 to May 2025

- Delivered backend features and testing infrastructure, reducing latency by 30% through algorithmic optimization and data-driven evaluation.